

Question 1:**a)****i.**

$$\left(\frac{-48}{229} \right)$$

Theorem 2.2(c), HB Chapter 10 p46

$$\left(\frac{-48}{229} \right) = \left(\frac{-1}{229} \right) \cdot \left(\frac{16}{229} \right) \cdot \left(\frac{3}{229} \right)$$

Calculating each term separately, using standard results

$$\left(\frac{-1}{229} \right) = 1$$

Theorem 2.2(e), HB Chapter 10 p46

$$\left(\frac{16}{229} \right) = 1$$

Theorem 2.2(b), HB Chapter 10 p46

$$\left(\frac{3}{229} \right) = 1$$

Proposition 4.7, HB Chapter 10 p46

Combining these results

$$\begin{aligned} \left(\frac{-48}{229} \right) &= 1 \cdot 1 \cdot 1 \\ &= 1 \end{aligned}$$

So -48 is a quadratic residue modulo 229**ii.**

$$\left(\frac{151}{43} \right)$$

$$\begin{aligned}\left(\frac{151}{43}\right) &= \left(\frac{22}{43}\right) \\ &= \left(\frac{2}{43}\right) \cdot \left(\frac{11}{43}\right)\end{aligned}$$

Theorem 2.2(c), HB Chapter 10 p46

Calculating each term separately

$$\left(\frac{2}{43}\right) = -1$$

Proposition 3.2, HB Chapter 10 p46

$$\left(\frac{11}{43}\right) = -\left(\frac{43}{11}\right)$$

Theorem 4.2, HB Chapter 10 p46

$$\begin{aligned}&= -\left(\frac{10}{11}\right) \\ &= -\left(\frac{2}{11}\right) \cdot \left(\frac{5}{11}\right)\end{aligned}$$

Theorem 2.2(c), HB Chapter 10 p46

Calculate each term separately

$$\left(\frac{2}{11}\right) = -1$$

Proposition 3.2, HB Chapter 10 p46

$$\left(\frac{5}{11}\right) = \left(\frac{11}{5}\right)$$

Theorem 4.2, HB Chapter 10 p46

$$\begin{aligned}&= \left(\frac{1}{5}\right) \\ \left(\frac{1}{5}\right) &= 1\end{aligned}$$

Theorem 2.2(b), HB Chapter 10 p46

Combining these results

$$\begin{aligned}\left(\frac{151}{43}\right) &= \left(\frac{22}{43}\right) = \left(\frac{2}{43}\right) \left(\frac{11}{43}\right) \\ \left(\frac{2}{43}\right) &= -1\end{aligned}$$

and

$$\left(\frac{11}{43}\right) = -\left(\frac{10}{11}\right) = -\left(\frac{2}{11}\right) \cdot \left(\frac{5}{11}\right)$$

$$\left(\frac{2}{11}\right) = -1$$

$$\left(\frac{5}{11}\right) = 1$$

So

$$\left(\frac{11}{43}\right) = -\left(\frac{2}{11}\right) \cdot \left(\frac{5}{11}\right) = -((-1)(1)) = 1$$

therefore

$$\left(\frac{151}{43}\right) = \left(\frac{2}{43}\right) \left(\frac{11}{43}\right) = (-1)(1) = -1$$

So 151 is a quadratic non-residue modulo 43